

The Cairo Spiny Mouse *Acomys cahirinus* Shows a Strong Affinity to the Mongolian Gerbil *Meriones unguiculatus*

Sergei I. Agulnik and Lee M. Silver

Department of Molecular Biology, Princeton University

The classification of the African spiny mice (genus *Acomys*) within the Muridae family of rodents has been fraught with controversy. Morphological data suggest a close affinity between this group and true old world mice of the genus *Mus*. However, the combined results of immunological, biochemical, and DNA melting studies suggest that spiny mice should not even be considered as members of the Murinae subfamily. To further elucidate the position of the spiny mice within the rodent phylogenetic tree, we performed a direct sequence comparison of a 583-nucleotide homologous region from the spiny mouse type species *Acomys cahirinus* and five other representative rodent species. Our results provide incontrovertible evidence to support the contention that the spiny mouse is more closely related to the Mongolian gerbil *Meriones unguiculatus* than it is to *Mus musculus*.

Introduction

The African spiny mice are a group of at least eight species of small rodents considered to be unique enough to be given the genus status *Acomys*. As their common name suggests, the *Acomys* group has been classified traditionally within the Murinae subfamily of old world mice and rats (Corbet and Hill 1991, p. 193; Wilson and Reeder 1993, p. 564). Furthermore, a shared complex derived character on the first upper molar within both *Mus* and *Acomys* (Jacobs 1978) and other shared morphological characters (Hutere, Lopez-Martinez, and Michaux 1988; Denys et al. 1992) have led some to suggest that African spiny mice are a recent offshoot of the old world mouse lineage. Upon analysis of immunological and biochemical data, however, a very different phylogenetic picture has emerged in which the spiny mice appear to be only distantly related to members of the genus *Mus* (Bonhomme et al. 1985; Sarich 1985; Wilson, Ochman and Prager, 1987). For mammals, this is the largest discrepancy known between a morphological and a molecular classification (Wilson, Ochman, and Prager 1987).

To better understand the position of spiny mice within the rodent phylogenetic tree, Catzeffis and his colleagues (Chevret et al. 1993) used the method of liquid-phase DNA-DNA hybridization with the non-highly repetitive fraction of DNA obtained from the type species *Acomys cahirinus* and samples derived from 20 other taxa—including true old world mice, gerbils, and hamsters—from various subfamilies within the family Muridae. By comparing the reduction in median melting temperature observed for DNA hybrids formed from

various pairs of species, these authors obtained strong evidence in confirmation of the placement of the spiny mice outside the clade defined by true old world mice and rats (subfamily Murinae) and within a monophyletic clade including the genera *Lophuromys* and *Uranomys* (both of which include animals referred to as brush-furred mice that are found exclusively in Africa). The data of Chevret et al. (1993) suggest that the *Acomys*/*Lophuromys*/*Uranomys* group is marginally more closely related to members of the Gerbillinae subfamily than to the Murinae subfamily.

In this report we describe the results of a direct sequence comparison that allows a further elucidation of the position of the spiny mice within the rodent phylogenetic tree and provides very strong evidence for a close relationship between the spiny mice and the Mongolian gerbil.

Materials and Methods

The phylogenetic analysis described in this report is based on DNA sequence from the fifth exon of the gene that encodes the kallikrein-binding protein (KBP). KBP is a liver-expressed Mr 60,000 glycoprotein that inhibits the action of kallikrein (Chai et al. 1991b). Synonyms for rat KBP include Spi-2.2, growth hormone-regulated protein and thyroid hormone-regulated protein (Chai et al. 1991b). The direct human homolog of rat KBP is known as α_1 -anti-chymotrypsin (Chai et al. 1991b). Other investigators have determined the complete nucleotide sequence of the KBP coding region from the genomes of the common rat *Rattus norvegicus* and the house mouse *Mus musculus* (Chai, Chao, and Chao 1991a; Chai et al. 1991b).

We used the fasta protocol from the Wisconsin GCG package to perform a screen for the presence of additional KBP-similar sequences in Genbank. This screen picked up a hamster KBP isolog (Accession no. M96650) that encodes a polypeptide named pregnancy

Key words: *Mus musculus*, *Acomys cahirinus*, *Meriones unguiculatus*, kallikrein-binding protein, rodent evolution.

Address for correspondence and reprints: Lee M. Silver, Lewis Thomas Laboratory, Department of Molecular Biology, Princeton University, Princeton, New Jersey 08540-1014. E-mail: LSILVER@Molbiol.princeton.edu.

	20	40	60	80	100
<i>A. cahirinus</i>	GTGG.TGCATAAGGCCATGATGGACGTGGCTGAGAAAGGCACAGAAGCAGCTGCTGCTACAGGAGTCAAATTTGCTCTATGTTGTCTAATAATAAGAAGT				
<i>M. unguiculatus</i>	GTGG.TGCACAAAGGCTGTGATGGACGTGGCTGAGAAAGGCACAGAAGCAGCTGCTGCTACAGGAGTCAAAGGAGTCCCTACTCTGTAGAAAAAGAGGAAGT				
<i>R. norvegicus</i>	GTGG.TCCACAAAGCTGTGCTGGATGTGGATGAGACAGGCACAGAAGGAGCCGCTGCCACAGCAGTACAGCAGCCCTAAAAAGTTTACGCCAACTATA				
<i>R. rattus</i>	GTGGGTCCACAAAGGCTGTGCTGAATGTGGATGAGACAGGCATGAGAGGAGCTGCTGCCCAATGGTCCACAAAGCCCTATATTATAGACCTCAAACCTATA				
<i>M. musculus</i>	ATGG.TCCACAAAGGCTGTGCTGGACGTGGCTGAGACAGGCACAGAAGGAGTTGCTGCCACAGGGGTCAATTTTCGATTCTTAGTAGA.....AGA				
<i>M. auratus</i>	GTGC.TCCACAAAGGCTGTGCTGGACGTGGGTGAGACGGGCACAGAGGCAGCGGCTGCCACAGGCCTCAGGATGATGGGATCAGCTCTAATTATGAACCTT				
	120	140	160	180	200
<i>A. cahirinus</i>	ATGATTGTGCGTTTTCAGACACCATTCCTCATGATTATCTGTGACACTAACAGTCAGATTATCCTCTTTATGTCAAAGTCACCAACCCCAAGTG....				
<i>M. unguiculatus</i>	ATGATTGTGCGTTTTCAGACACCATTCCTCATGATTATCTGTGACACTAACAGTCAGATTATCCTCTTTATGTCAAAGTCACCAACCCCAAGTTG....				
<i>R. norvegicus</i>	CCTCTTCTCAATTTCAACCGGCCATTCTGCTGGTTATCACTGACAATAAGTTCAGTCTGCTCTTTATGGGCAAGTCACCAACCCCATCTGAGTCT				
<i>R. rattus</i>	...AATGTGAATTTCAACCGGCCATTCTGCTGGTTATCACTGACAATAAGTTCAGTCTGCTCTTTATGGGCAAGTCACCAACCCCATCTGAGTCT				
<i>M. musculus</i>	ACCACTGTGCTTTCAGACGACATTCTGATGGTCATCTCTCACACAGATGTTCACTACTCTCTTTATAGCCAAATCACTCACCCCTAAGAGAGCCT				
<i>M. auratus</i>	CTGAGACTGCGTTTTCAGACGACATTCTGATGGTCATCTCTATAACCAAACTCACTCAGACTCCTCTTTATGGGCAAGTCACCAATCCCAAGCAAACT				
	220	240	260	280	300
<i>A. cahirinus</i>	TGACTCCTCTGCCCAAGTCCGAGGAG.GAGCCTGCATGTGATTCCCGTGT.GCCCTCTGCAATCATGCTCTGATTGGCTGCT				
<i>M. unguiculatus</i>	TGACTCCTTGGCCAGGAATCAGGAGGAGCCTGCATGTGATTCTATGGGTGGCTCCTGCCATTCTGCTCTGATTGGCTGCT				
<i>R. norvegicus</i>	GAAAGTCCCCAAATCTGACAATTCTGCCAGGATCCTGGAACAGAGCCTGGATGCTGATCTCTGTATATGCCCTGACATTACATGCTCTGATTGGCTATT				
<i>R. rattus</i>	GAAAGTCCCCAAATCTGACAGTTCTGCCAGGATCCTGGAACAGAGCCTGGATGCTGATCTCTGTATATGCCCTGACATTACATGCTCTGATTGGCTATT				
<i>M. musculus</i>	AAAACTGTCCAAGTTCTGAATATTCTGCCAGGACTCTGAGATGCAACCTATATGTGGATCTCTCTTTATGCTGCAATCAAGCTCTGATTGGCTGCT				
<i>M. auratus</i>	AGAACATCCCAAGTCTGAGGCTCTTCCAGGAGCCAGGATTAATCCTGTATGAGAGTATCTATCTGCATTTTCTGCTCCATGCTCTGATTGGCTGCT				
	320	340	360	380	400
<i>A. cahirinus</i>	TTCTAGCTG.CTTAGA.AAGTGACAGGGAGTATTTCTGTGGCTCCACAGGCACAGAAGCCTTCAGA.TGTCAGTGTGAGGCTCTCAGGCTCTCTGGGA				
<i>M. unguiculatus</i>	TTCTAGCTGCTTAGA.AAGTGACAGGGAGTATTTCTATGGCTCCACAGGCACAGAAGCCTTTGGA.TGTCAGTGTGAGGCTCTCAGGCTCTCTGGGA				
<i>R. norvegicus</i>	GCAAAGTTGGCTTAGACCAAGTGCATCAATCTCTATGGCTCCCATGTGCATTGGAGCCTTTGGATTGTGAGTGTGAGGCTCTCAGGCTCTCTGGGA				
<i>R. rattus</i>	GCAAAGTTGGCTTAGACCAAGGACATCGAC.ATCTCTATGGCTCCCATGTGCACCAAGGCTTTGGATTGTGAGTGTGAGGCTCTCAGGCTCTCTGGGA				
<i>M. musculus</i>	GCAAAGTTGGCTTAGA.CAGCGACATTGACTATCTCGTGGCTCCACATGCACAGAAGCCTTTGGACTATCTGTGAGGCTCTCGGCTCTCTGGGA				
<i>M. auratus</i>	GCATGCTGAGTTGGACAA.TGACAGTAACCTATGACTGCCACATGCACAGAGTCTGGGACAGTCACTGTGAGGCTCTCAGGCTCTCTGGGA				
	420	440	460	480	500
<i>A. cahirinus</i>	AACCTACACATGTTTCTCAACCTGGATTCTGTCTTTATTCTTCTCTCACTGCTGACTCCACTTTCTGTGCTGTTCCCGAGACCTAGGCACTGA.AAGTA				
<i>M. unguiculatus</i>	AACCTGCACATGTTTCTCAACCTGGATTCTGTCTTTATTCTTCTCTCACTGCTGACTCCACTTTCTGTGCTGTTCCCTAGACCTAGGCACTGA.AAGTA				
<i>R. norvegicus</i>	CATCTACACATGTTTCTGAACCTGGAATCTTTCTTTATTCTTCTCTCCCTGGTGACTCCTCTTTCTGTGTTTCATACCCCAACCAAGCCATTGATA....				
<i>R. rattus</i>	CATCTACACATGTTTCTGAACCTGGAATCTTTCTTTATTCTTCTCTCCCTGGTGACTCCTCTTTCTGTGTTTCATACCCCAACCAAGCCATTGATA....				
<i>M. musculus</i>	CATCTGTCAGCTTTCTGAACCTGGAATCTGCTTTATTCTTCTCTCCCTGGTGACTCCTCTTTATGATCTGTATCCCAACCTAGGCACTTGGCAATTA				
<i>M. auratus</i>	CACTAGCTCATATTCTGAAGCTGAAATCTATCCTTGTGGCACTCCTTCCCGCACTATTTTATCTGCTCTACACAAAGCCTGGGCCCTGGAAAGTA				
	520	540	560	580	
<i>A. cahirinus</i>	CTGCAGTCCCAGCACAAGTTCTGGTCACATCTGCCAT.AGCTTCCATAGTCTT.CTGGGACCATGCAAGCTTAAAGGCCAA.C				
<i>M. unguiculatus</i>	CTGCAGT.CCAGCACAAGTTCTGGTCACATCTGCCAT.AGCTTCCATAGTCTT.CTGGGACCATGCAAGCTTAAAGGCCAA.C				
<i>R. norvegicus</i>	AGTCCCAGTAAAGGTTCTGAGAACATCTGTTTTCAGTCTCCAGAGTCTTGATGAAACTAGGCTAGCTC.TGGGCCAACCC				
<i>R. rattus</i>	AGTCCCAGTAAAGATCTGATTACATTTGTTTTCAGTCTCCACACTCTTGATGGAAGTAAAGCTAGCTCCTGGGCCAA.C				
<i>M. musculus</i>	ACAGAGTCCAGTATAGGCTCTGAGCACATCTGCT.....TGACAGTCTTGATGGGACCATGAAAGCCTCTTTGCCAATC				
<i>M. auratus</i>	GGCTCACCCCTGTCTAGGCTCTTGTATGTCTACATTCAGCTTCCACAGCCTTCATGGAATCATGCAAACTTATAGGTGAACC				

FIG. 1.—Homologous DNA sequences from the fifth exon of the kallikrein-binding protein gene. The DNA sequence starts at the beginning of the fifth exon, which is the last exon of the gene. The open reading frame of each sequence is indicated within the box.

protein 60 kDa—with the same size and expression pattern as KBP. Strong sequence similarity with the previously characterized KBP genes suggests that this hamster gene is a direct KBP ortholog. However, this assignment is not required for the use of this sequence as an outgroup in the phylogenetic analysis reported here.

Conserved stretches of sequence from the rat and mouse KBP homologs were used to develop a pair of primers (5'-TCTCCACTGAGTGATGTGTT-3') and (5'-GTCCTGGGCCTGATTCTCATG-3') for the polymerase chain reaction (PCR) amplification and cloning of a ~600-bp portion of the fifth exon of the KBP gene from three additional species: the black rat (*Rattus rattus*), the Cairo spiny mouse (*A. cahirinus*), and the common Mongolian gerbil (*Meriones unguiculatus*). A sample of *A. cahirinus* DNA was kindly provided by François Catzeflis (Montpellier, France). A Mongolian gerbil was obtained from a local commercial supplier and used for the preparation of liver DNA according to standard pro-

cedures. A sample of *R. rattus* DNA was kindly provided by Priscilla Tucker (Ann Arbor, Michigan).

The amplification conditions were as follows: 94°C, 1 min; 55°C, 1.5 min; 72°C, 2 min repeated for 35 cycles. Amplified products were examined by gel electrophoresis and ethidium bromide staining. In all cases, only a single band of the expected size was observed. PCR products were purified using a PCR purification kit (Qiagen) and cloned with a TA-cloning kit (Invitrogen). Sequencing was performed with the Sequenase kit (U.S. Biochemicals) and the Circumvent kit (New England Biolabs). All sequences were confirmed in at least two independently derived clones. Phylogenetic analysis was performed using the PHYLIP program version 3.52c (Felsenstein 1991) on a Sparc workstation, the Wisconsin GCG package (Genetics Computer Group 1989) on a VAX minicomputer, and the MacClade program (Maddison and Maddison 1992) on a Macintosh computer.

Results

The Pileup protocol from the Wisconsin package was used to line up the six KBP sequences as shown in figure 1. The 583-nt homologous region starts from the beginning of the fifth exon and continues through the coding region into the 3'-untranslated region. The portion of each sequence that codes for polypeptide is boxed in. The KBP stop codon occurs in four different positions among the six sequences. The coding regions of the two rat homologs ends at nucleotide 193, the coding region of the *Acomys* homolog is extended further by one codon, and the coding regions of the mouse and hamster homologs is extended further still by one codon. Small changes in nucleotide sequence cause an extension of the gerbil sequence by 35 additional codons beyond the mouse sequence.

A phylogenetic analysis based on maximum likelihood was performed with the PHYLIP program of Felsenstein (1991). The golden hamster sequence was chosen as an outgroup based on traditional systematics (Chevret et al. 1993). Five hundred bootstrap replicates were performed and the majority-rule strict consensus tree shown in figure 2 was obtained at a frequency of 98.6%. The only other tree obtained in this analysis (at a frequency of 1.4%) differed from the one shown in figure 2 through the placement of the mouse in association with the hamster lineage. A second, independent phylogenetic analysis of the same data set was based on 500 bootstrap replicates with maximum parsimony. Once again, the *Acomys* sequence was found to associate with the gerbil sequence, and not the mouse or rat sequences, in all 500 trees.

The PHYLIP-derived consensus tree and the associated sequence data set were entered into the MacClade program (Maddison and Maddison 1992) to calculate estimates for the length of each branch along the tree. Three different estimates of branch length were obtained: a minimum estimate based on unambiguous changes unique to each branch; a maximum estimate based on all possible changes that may have occurred along each branch; and an average estimate that gives the most likely set of changes that occurred along each branch. The relative branch lengths obtained with all three estimation protocols are displayed in figure 2.

Discussion

The phylogenetic analysis presented in this report provides strong support for the phylogenetic view suggested by previous molecular studies: spiny mice do not appear to be members of the Old World mouse clade. Rather, our data provide strong support for the branching of spiny mice from the line leading to the gerbil subfamily Gerbillinae.

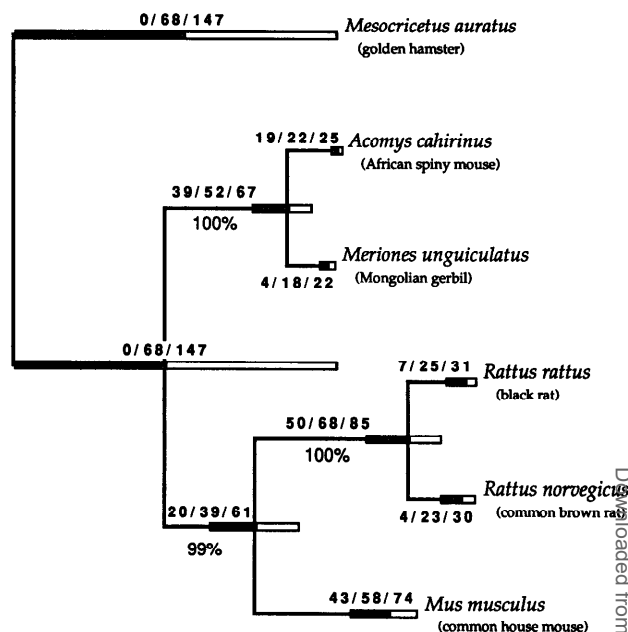


FIG. 2.—Phylogenetic tree based on the kallikrein-binding protein gene. The topology of the tree was determined with PHYLIP program as described in the text. The common name for each species is indicated beneath the formal name. The *Mesocricetus auratus* sequence was chosen as the outgroup. Three estimates of branch length determined with the MacClade program are indicated and represented proportionally along each branch. The first number is the minimum length based on “unambiguous” changes; this length is represented by the solid portion of each branch. The second number is the average branch length based on the most likely set of changes; this length is represented by the portion of each branch that extends through the shaded part of the box. The third number is the maximum branch length based on all possible changes; this number is represented by the complete extension of each branch. Indicated below each ancestral branch is the percentage of bootstrap runs that yielded the same set of species contained within.

If one assumes a molecular clock for the KBP gene that ticks a constant rate among all of the rodent lineages examined in this study, the Cairo spiny mouse and the Mongolian gerbil would appear to be as closely related to each other as two species—*R. rattus* and *R. norvegicus*—within the single genus *Rattus*. If one assumes a time of divergence for rats and true mice between 15 and 30 myr bp (Wilson, Ochman, and Prager 1987), a corresponding estimate for the time of divergence of spiny mice and gerbils would be 5–10 myr bp. Confirming evidence in support of the close relationship between the Cairo spiny mouse and the Mongolian gerbil has been obtained in a recent study of partial sequences from a second nuclear locus named TCTE1 (S.A. and L.M.S., unpublished data). Further studies with additional loci and with additional species types in the subfamily Gerbillinae will be required to more accurately determine the relative location of the spiny mice within this clade.

Acknowledgments

We wish to thank François Catzefflis for his kind gift of purified *A. cahirinus* DNA, and Priscilla Tucker for her kind gift of *R. rattus* DNA samples. We thank an anonymous reviewer for comments that led to a reassessment of the data presented in this manuscript. This work was supported by a grant from the NIH to L.M.S.

LITERATURE CITED

- BONHOMME, F., D. ISKANDER, L. THALER, and F. PETTER. 1985. Electromorphs and phylogeny in muroid rodents. Pp. 671–684 in W. P. LUCKETT, and J. L. HARTENBERGER, eds. *Evolutionary relationships among rodents: a multidisciplinary approach*. Plenum, New York.
- CHAI, K. X., J. CHAO, and L. CHAO. 1991a. Molecular cloning and sequence analysis of the mouse kallikrein-binding protein gene. *Biochem. Biophys. Acta* **1129**:127–130.
- CHAI, K. X., J.-X. MA, S. R. MURRAY, J. CHAO, and L. CHAO. 1991b. Molecular cloning and analysis of rat kallikrein-binding protein gene. *J. Biol. Chem.* **226**:16029–16036.
- CHEVRET, P., C. DENYS, J.-J. JAEGER, J. MICHAUX, and F. M. CATZEFLIS. 1993. Molecular evidence that the spiny mouse (*Acomys*) is more closely related to gerbils (*Gerbillinae*) than to true mice (*Murinae*). *Proc. Natl. Acad. Sci. USA* **90**:3433–3436.
- CORBET, G. B., and J. E. HILL. 1991. *A world list of mammalian species*. Oxford University Press, New York.
- DENYS, C., J. MICHAUX, F. PETTER, J.-P. AGUILAR, and J. J. JAEGER. 1992. Molar morphology as a clue to the phylogenetic relationship of *Acomys* to the *Murinae*. *Isr. J. Zool.* **38**:253–262.
- FELSENSTEIN, J. 1991. *PHYLIP (Phylogeny Inference Package)*. Univ. Washington, Seattle.
- GENETICS COMPUTER GROUP. 1989. *The Wisconsin GCG package*. Maryland Biotechnology Institute, Madison, Wis.
- HUTTERE, R., N. LOPEZ-MARTINEZ, and J. MICHAUX. 1988. A new rodent from the Quaternary deposits of the Canary islands and its relationship with Neogene and recent murids of Europe and Africa. *Paleovertebrata* **18**:241–262.
- JACOBS, L. L. 1978. Fossil rodents (*Rhizomyidae* and *Muridae*) from Neogene Siwalik deposits, Pakistan. *Bull. Mus. North. Ariz. Press ser.* **52**:1–103.
- MADDISON, W. P., and D. R. MADDISON. 1992. *MacClade: analysis of phylogeny and character evolution*. Sinauer, Sunderland, Mass.
- SARICH, V. M. 1985. Rodent macromolecular systematics. Pp. 423–452 in W. P. LUCKETT, and J.-L. HARTENBERGER, eds. *Evolutionary relationships among rodents: a multidisciplinary approach*. Plenum, New York.
- WILSON, A. C., H. OCHMAN, and E. M. PRAGER. 1987. Molecular time scale for evolution. *Trends Genet.* **3**:241–247.
- WILSON, D. E., and D. M. REEDER, eds. 1993. *Mammal species of the world: a taxonomic and geographic reference*. Smithsonian Institution Press, Washington.
- JEFFREY R. POWELL, reviewing editor

Accepted July 18, 1995